Managing Disks, RAM Usage, and Swap on Linux

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1 Why This Session Matters

Before we touch commands, understand this:

- Your **disk** stores data permanently.
- Your **RAM** is fast temporary memory.
- **Swap** is emergency backup memory.
- **zram** is compressed RAM swap (smart trick for low-memory systems).

If you understand this session, you'll:

- Mount extra drives safely
- Fix broken boot issues
- Add a second disk like a pro
- Understand RAM pressure
- Boost low-RAM machines using zram

> This is not theory. This is real Linux power.

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2 Checking If `zram-tools` Is Installed

On Debian/Ubuntu/Mint, zram is handled by:

```
$ sudo apt install zram-tools
```

To check if it's installed:

```
$ dpkg -l | grep zram
```

To check if it's active:

```
$ swapon --show
```

If you see something like `/dev/zram0`, it's working.

To check service status:

```
$ systemctl status zramswap
```

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3 🖴 Understanding Your Disks

Let's meet your disks.

🌲 `lsblk` — The Disk Tree Viewer

```
$ lsblk
```

Example output:

```
sda
  sda1
  sda2
  sda3
sdb
  sdb1
```

What it shows:

- Disk names (`sda`, `sdb`)
- Partitions (`sda1`, `sda2`)

- Mount points
- Size
- Type

More detailed output:

```
$ lsblk -f
```

This shows:

- Filesystem type (ext4, vfat, ntfs)
- UUID
- Mountpoint

Use this before and after plugging a USB.

`blkid` — UUID Finder

```
$ sudo blkid
```

It shows something like:

```
/dev/sda1: UUID="1A2B-3C4D" TYPE="vfat"
```

Why important?
Because in `/etc/fstab`, we use **UUID**, not `/dev/sda1`.
Because device names can change on boot. UUID never lies.

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4 Mounting and Unmounting Drives

Unlike Windows, Linux does not auto-use disks unless mounted.

Manual Mount

Step 1 — Create mount point:

```
$ sudo mkdir /mnt/mydrive
```

Step 2 — Mount:

```
$ sudo mount /dev/sdb1 /mnt/mydrive
```

Step 3 — Check:

```
$ ls /mnt/mydrive
```

Unmount

```
$ sudo umount /mnt/mydrive
```

OR

```
$ sudo umount /dev/sdb1
```

If busy error:

```
$ lsof | grep sdb1
```

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5 Making Mount Permanent — Editing `/etc/fstab`

This file controls what mounts at boot.
Open it:

```
$ sudo nano /etc/fstab
```

Example entry:

```
UUID=1a2b3c4d-xxxx-xxxx-xxxx-xxxxxxxxxxxx /mnt/mydrive ext4 defaults 0 2
```

Columns explained:

Column	Meaning
UUID	Disk identifier
Mount point	Where it appears
Filesystem	ext4, ntfs, vfat
Options	defaults, noatime
Dump	usually 0
fsck order	usually 2

Test Before Reboot

After editing:

```
$ sudo mount -a
```

✔ If no errors → safe to reboot.
⚠ If error → FIX IT before rebooting.

Practical Example: Adding Second Drive

→ `mmbesar-2nd-drive`

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6 Monitoring RAM Usage

RAM = short-term memory.

`free`

```
$ free -h
```

Example:

```
 total used free shared buff/cache available
Mem:  7.6G 3.1G 1.2G 500M 3.3G 3.8G
Swap:  2.0G 0B 2.0G
```

Important columns:

- **used** → currently used
- **available** → real usable memory
- **buff/cache** → Linux using RAM smartly

⚠ Linux uses RAM aggressively. That's GOOD.
Unused RAM = wasted RAM.

Real-time View

```
$ watch -n 1 free -h
```

Updates every second.

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7 ↔ Understanding Swap

Swap = disk used as overflow memory.
When RAM fills up → inactive pages move to swap.
But swap is slower than RAM.
Check swap:

```
$ swapon --show
```

or

```
$ cat /proc/swaps
```

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8 Creating a Swap File (Manual Way)

If no swap exists:
Step 1:

```
$ sudo fallocate -l 2G /swapfile
```

Step 2:

```
$ sudo chmod 600 /swapfile
```

Step 3:

```
$ sudo mkswap /swapfile
```

Step 4:

```
$ sudo swapon /swapfile
```

Make permanent:
Add to `/etc/fstab`:

```
/swapfile none swap sw 0 0
```

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9 ⁝⁝ zram — Compressed RAM Swap

zram creates compressed swap in RAM.
Instead of writing to disk:

- It compresses pages
- Stores them in RAM
- Faster than disk swap

Great for:

- 4GB RAM laptops
- Old machines
- Low-resource VPS

Install on Debian/Ubuntu/Mint

```
$ sudo apt install zram-tools
```

Check config:

```
$ cat /etc/default/zramswap
```

You can adjust:

```
PERCENT=70
```

Meaning: use 70% of RAM for zram.
You can change it but it is recommended to adjust it to [70%-80%].

Restart:

```
$ sudo systemctl restart zramswap
```

Check:

```
$ swapon --show
```

You should see `/dev/zram0`.

Checkout This Practical Example in the link below:
→ `mmbesar-RAM-SWAP-ZRAM`

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10 ⚖️ Swap vs zram Comparison

Feature	Disk Swap	zram
Speed	Slow	Faster
Uses disk	Yes	No
Best for	Large systems	Low RAM
Compression	No	Yes

Best practice for laptops:

- Use zram
- Keep small disk swap

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11 🛠 Troubleshooting Boot Failures (fstab Mistakes)

If system won't boot:

1. Boot into recovery mode
2. Mount root
3. Edit `/etc/fstab`
4. Fix or comment broken line

> Pro tip:
> Use this option in fstab for external drives:

```
nofail
```

> So system doesn't panic if drive missing.

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12 ☰ Useful Extra Commands

Check disk usage:

```
$ df -h
```

> Check disk space by folder:

```
$ du -sh *
```

> See mount points:

```
$ mount | grep sdb
```

🧪 Practical Lab Exercises

1. **Exercise 1:** Plug USB → identify it using `lsblk`.
2. **Exercise 2:** Mount it manually in `/mnt/testusb`.
3. **Exercise 3:** Make it permanent in fstab.

4. **Exercise 4:** Create 1GB swap file.
5. **Exercise 5:** Install zram and compare swap usage.

Suggested Video References

- `https://www.youtube.com/watch?v=yWuHI7uoftY`
- `https://youtu.be/CPvJvY79vEw`

(You should watch after hands-on practice.)

Final Advice

Disk and memory management is not optional knowledge.
It's what separates:

"Linux user"
↓
"Linux operator"

- Break things in a VM.
- Test everything with `mount -a`.
- Never reboot blindly after editing fstab.

You're building serious Linux skills now.

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